Sean Lee

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EDUCATION

Keck School of Medicine, University of Southern California — **Los Angeles, CA**
Doctor of Medicine Candidate — Expected Graduation May 2028
Honors: Family Medicine | High Pass: Internal Medicine, Surgery, Pediatrics, Neurology, Psychiatry

University of Southern California
Bachelor of Science, Human Biology | Minor in Healthcare Studies — 2019 - 2022

PEER-REVIEWED PUBLICATIONS

Cullaro G, Lai JC, Liu T, Gomez ME, Simonetto DA, **Lee S**, Lee BP, Allegretti AS, Mazumder NR, Patidar KR, Verna EC. Machine learning-derived clinical phenotypes of ascites in cirrhosis: A multi-center latent class analysis with external validation. Liver Transpl. 2026. doi:10.1097/LVT.0000000000000942.

Meremikwu K, **Lee S**, Tang B, Wilson ML, Killu K. Diagnostic accuracy of lung ultrasound for pneumonia in adult intensive care unit patients: A systematic review and meta-analysis. Cureus. 2026;18(9):e115858. doi:10.7759/cureus.115858.

Munoz XC, Elshafei M, Pouwels S, **Lee S**, Eini JI, Lee C, Kerendian E, Ichiuji B, Akhnoukh S, Cironi K, Mihalsky KP, Kuiper JJ, Putnam LR, Wong HJ, Galvin KM, Abel SA, Samakar K, Shiraga S, Lipham JC, Houghton CC. Achieving physiologic distensibility targets frequently requires modification of standard cruroplasty during anti-reflux surgery. Surg Endosc. 2026. doi:10.1007/s00464-026-13365-5.

MANUSCRIPTS IN PREPARATION / UNDER REVIEW

Elshafei M, Pouwels S, Putnam L, Munoz X, Eini JI, **Lee S**, Lipham JC, Houghton C. Implementation of a standardized protocol for EndoFLIP utilization alters intraoperative hiatal closure: a proof-of-concept study. (Submitted)

Lee S, Ameri H, et al. Changes in ophthalmic care and 3-year outcomes among patients with proliferative diabetic retinopathy: a national TriNetX cohort study.

Suh A, **Lee S**, Szeto J, et al. Beyond Earth-based care: a proposed ophthalmic surgical training framework for crew medical officers during long-duration spaceflight.

Suh A, **Lee S**, Szeto J, Kumar D, Ong J, Berdahl J, Lee AG. The ocular–behavioral health axis in spaceflight: interconnections of two red risks on the mission to Mars.

Banashefski B, Achilles D, **Lee S**, Lin M, Feldman C. Ocular disease burden and diagnostic and treatment patterns in street-based health clinics in the United States.

[First author], [co-authors], **Lee S**, [co-authors]. A cystatin C–augmented MELD score for 90-day waitlist mortality: development and validation in a multicenter US liver transplant consortium.

Lee S, Cironi K, Galvin K, Lipham JC, et al. Comparative pathologic outcomes of FLOT vs CROSS neoadjuvant therapy for resectable esophageal cancer: a single-institution retrospective series.

POSTER & ORAL PRESENTATIONS

Lee S, Lipham JC, et al. Comparative pathologic outcomes of FLOT vs CROSS neoadjuvant therapy for resectable esophageal cancer: a single-institution retrospective series. Poster presented at: American Foregut Society Annual Meeting; September 24–26, 2026; National Harbor, MD.

Banashefski B, Achilles D, **Lee S**, Lin M, Feldman C. Ocular disease burden and diagnostic and treatment patterns in street-based health clinics in the United States. Projected Feb 2027

Lee S, Ameri H, et al. Changes in ophthalmic care and 3-year outcomes among patients with proliferative diabetic retinopathy: a national TriNetX cohort study: ARVO Projected 2027

RESEARCH

USC Keck School of Medicine — **Los Angeles, CA**
Department of Ophthalmology, Roski — May 2026 - Present
Research Assistant — Dr. Hossein Ameri

- Independently design retrospective cohorts and conduct statistical analyses using the TriNetX federated research network to investigate longitudinal trends in diabetic retinopathy management, treatment utilization, vision-threatening complications, and

healthcare resource utilization.
• Utilize Optos ultra-widefield (UWF) retinal imaging to characterize retinal findings in patients with inherited retinal diseases and support the development of imaging-based research.
• Contribute to an artificial intelligence–based retinal imaging study involving segmentation and quantitative analysis of retinal vasculature in ultra-widefield images of patients with retinitis pigmentosa (RP).
• Gain hands-on experience in rodent-based ophthalmic wet-lab research, with involvement in preclinical CRISPR-mediated gene knockout and anti-VEGF investigations utilizing subretinal and intravitreal injections in mouse models.

Research Assistant — Dr. Alexander Suh May 2026 - Present
• Conducted literature review and contributed to manuscript development for a narrative review examining the Ocular-Behavioral Health Axis during long-duration spaceflight, including interactions among SANS, cognitive load, sleep deprivation, and psychiatric sequelae.
• Serve as primary student author developing a proposed ophthalmic procedural training framework for Crew Medical Officers during exploration-class spaceflight missions, incorporating emergency procedure prioritization, autonomous surgical capabilities, and communication delays.

Department of General Surgery (GLS/MIS)
Research Assistant — Dr. John Lipham, Dr. Luke Putnam Jan 2026 - Present
• Conducted retrospective chart review and data collection evaluating the impact of intraoperative platelet-rich plasma (PRP) injection on postoperative outcomes following hiatal hernia repair.
• Contributed to a published study investigating standardized intraoperative EndoFLIP utilization and its impact on hiatal closure techniques during antireflux surgery, including clinical data collection and manuscript development.
• Prepared manuscript results and contributed to a published study evaluating physiologic distensibility targets, intraoperative cruroplasty modifications, and postoperative outcomes following antireflux surgery.

Department of Pulmonary & Critical Care
Research Assistant — Dr. Keith Killu Nov 2025 - May 2026
• Contributed to a published systematic review and meta-analysis evaluating the diagnostic accuracy of lung ultrasound for pneumonia, conducting literature screening, data extraction, methodological quality assessment, and quantitative evidence synthesis.
• Independently developed a retrospective research proposal investigating clinical and laboratory predictors of CT utilization in hospitalized pneumonia patients, including hypothesis formulation, cohort design, variable selection, and preparation of IRB submission materials.

Department of Transplant Hepatology
Research Assistant — Dr. Brian Lee, Dr. Giuseppe Cullaro Sep 2024 - May 2026
• Conducted longitudinal chart review and clinical data extraction for 150+ liver transplant patients, contributing to a multicenter study evaluating Cystatin C–augmented MELD scoring for improved mortality risk prediction.
• Contributed to a published multicenter study investigating machine learning–derived clinical phenotypes of ascites in cirrhosis through patient chart review, data extraction, and manuscript development.

Department of Emergency Medicine
Research Assistant — Dr. Alexander Sapin April 2026 - Oct 2026
• Contributed to a quality improvement initiative aimed at developing an automated flagging system to identify ED patients at risk for prolonged length of stay (LOS >21 days) to facilitate early disposition planning.
• Performed data collection and analysis of admission diagnostic codes for patients meeting LOS threshold, forming the evidential basis for future flag criteria development.

USC Alfred E. Mann School of Pharmacy **Los Angeles, CA**
Department of Pharmacology and Pharmaceutical Sciences June 2020 - June 2022
Research Assistant — Dr. Yong (Tiger) Zhang
• Conducted weekly literature reviews screening 200+ papers to identify novel antibody candidates for smart-exosome-based anticancer drug delivery research.
• Performed protein and DNA purification, cell culture maintenance, and preparation of Antibody-Drug Conjugate specimens in support of CD36-targeted cancer antibody studies.

OPHTHALMOLOGY CLINICAL EXPERIENCE

USC Keck School of Medicine **Los Angeles, CA**
Department of Ophthalmology, Roski Aug 2026
Ophthalmology Clinical Elective
• Performed supervised ophthalmic examinations, including visual acuity assessment, slit-lamp biomicroscopy, and fundoscopic examination.
• Evaluated patients in outpatient ophthalmology clinics, presented clinical findings to attending ophthalmologists, and participated in discussions of differential diagnoses and management plans.

PROJECTS

Rounds (Independent Project) May 2026 - Present
• Designed and developed *Rounds*, a cross-platform mobile app (iOS/Android) in React Native to reduce cognitive load during

clinical rotations, featuring H&P and SOAP note templates, patient task tracking, lab result organization with intelligent auto-fill, A/P templating, clinical timers, and shift signout checklists; no patient data stored.
[*https://apps.apple.com/app/rounds-qf5429/id6769269478*]

Note-Helper, an AI-Powered Clinical Documentation Tool (Independent Project) April 2026
• Built a HIPAA-compliant web tool using keyword recognition and speech dictation to generate structured clinical notes, standardizing documentation output across encounter types without retaining patient data.
[https://github.com/infer321/note-helper]

Antibiotic Stewardship Tool (Independent Project) May 2026
• Developed a publicly deployed web-based prescribing reference covering 30+ infection types across 9 organ systems, with dynamic adjustments for penicillin allergy and MRSA risk; delivers setting-appropriate regimens (outpatient/floor/ICU) aligned with current IDSA/ASHP guidelines in 3 clicks.
[https://github.com/infer321/AntibioticStewardshipTool]

Internal Medicine Clinical Scoring Calculator (Independent Project) May 2026
• Built a mobile-optimized, offline-capable web calculator consolidating 15 validated scoring tools (HEART, CURB-65, MELD-Na, Wells PE/DVT, SOFA, APACHE II, and others) with real-time interpretation and evidence-based clinical guidance.
[https://github.com/infer321/InternalMedCalc]

PERTINENT EMPLOYMENT HISTORY

AHMC: Garfield Medical Center **Monterey Park, CA**
Patient Liaison | Project Manager May 2022 - May 2023
• Collaborated with physicians and nursing staff to address urgent patient needs and served as a Cantonese–Mandarin–English medical interpreter, facilitating communication during clinical encounters, discharge planning, and informed consent.
• Analyzed HCAHPS, Leapfrog, and emergency department throughput metrics across a 9-hospital system, synthesizing performance data to support quality improvement initiatives and executive decision-making.
• Conducted literature reviews of emerging COVID-19 evidence for biweekly grand rounds and supported senior clinical and administrative leadership through meeting documentation and coordination of operational initiatives.

LEADERSHIP/VOLUNTEER EXPERIENCE

USC Keck School of Medicine **Los Angeles, CA**
Office of Student Affairs | *Student Affairs Representative* Aug 2023 - Aug 2025
• Represented the medical student body in discussions with institutional leadership, advocating for student needs and improvements in campus resources, housing equity, and student well-being.
• Collaborated with administrators to identify and implement initiatives addressing facility improvements, mental health resource accessibility, and student engagement.
Gastroenterology Student Interest Group | *Vice President* Aug 2023 - Mar 2025
• Spearheaded programming initiatives by coordinating with faculty and clinical speakers to curate lunch talk series, delivering career-relevant insights to students pursuing medicine.
• Led targeted outreach to identify and compile a diverse roster of research-active faculty, connecting students with mentorship and research opportunities across a broad range of subspecialties and interests.
USC Photography Club | *President* | *Events Director* Aug 2020 – Aug 2023
• Led a 15-person executive team to organize events, retreats, and workshops for 2,000+ members, overseeing event planning, marketing, and logistics operations.

Dmanisi Archaeological Field School **Dmanisi, Georgia**
Field Researcher Jun 2019 - Aug 2019
• Participated in active excavation at the Dmanisi site, one of the most significant *Homo erectus* fossil sites outside Africa, gaining hands-on exposure to paleoanthropological methods and human evolutionary history.
• Attended lectures on hominin morphology, species differentiation, and the anthropological significance of early human migration.

Alhambra Hospital **Alhambra, CA**
Emergency Medicine Shadowing | Volunteer July 2020 – Dec 2022
• Observed clinical decision-making in high-acuity emergency settings across diverse patient populations, and assisted nursing staff with patient care, vitals, and unit operations.

Connect Tech Network **San Francisco, CA**
Computer and Technological Tutor | Volunteer July 2020 – Jun 2022
• Taught digital literacy (Zoom, Google tools) in Cantonese to elderly clients during COVID-19 to improve accessibility of remote services.

Academic Mentor **Alhambra, CA**
Independent Mentor Sept 2022 – Present
• Provided longitudinal academic mentorship to a student across grades 6–10, focusing on science, mathematics, and evidence-based study strategies. Emphasized stepwise problem-solving, self-directed learning habits, and metacognitive skill development.

TECHNICAL & RESEARCH SKILLS

Ophthalmic Imaging: Optos ultra-widefield (UWF) retinal imaging, retinal vessel segmentation, quantitative retinal vascular analysis.
Clinical Research: TriNetX, retrospective cohort design, propensity score matching, Kaplan–Meier survival analysis, hazard ratios and risk ratios, longitudinal outcomes analysis, systematic reviews, meta-analysis.
Software Development: React Native, clinical decision-support applications, AI-assisted clinical documentation tools.

CERTIFICATES

Quality Improvement Certificate | USC Keck School of Medicine **2026**
- Completed formal training in QI methodology, including process improvement frameworks and patient safety principles applicable to clinical practice.

Certificate of Merit Piano, Level 10 | Music Teachers' Association of California **2018**
- Achieved the highest level of the statewide classical piano assessment, demonstrating years of disciplined technical and musical development

LANGUAGES

English – Native
Chinese Cantonese – Conversational proficiency
Chinese Mandarin – Intermediate (able to take patient history and perform physical exam)
Spanish – Intermediate (able to take patient history and perform physical exam)

PROFESSIONAL INTERESTS

Piano (Performed Chopin Ballade No. 1, 2019 MTAC San Diego Chopin Competition)
Weightlifting, Cycling, Running, Photography